

CAMPUS NAVIGATION MOBILE APPLICATION

SCREEN 1: SPLASH / WELCOME SCREEN

The Welcome Screen is the first interface displayed when the user launches the Campus Navigation application.

The purpose of this screen is to introduce the application and provide a clear starting point for the user.

Main UI Elements

- Campus Navigation logo.
- Application name.
- Short tagline.
- Campus-related visual.
- Primary Get Started button.

Sample Interface Content

CampusNav

Find Your Way Around Campus

Navigate your campus easily and reach your destination faster.

GET STARTED

Design Considerations

The screen uses a clean layout with minimal content. The application name and tagline immediately communicate the purpose of the application. The primary button is positioned prominently so that users can easily begin using the application.

User Interaction

The user taps the Get Started button to proceed to the Home / Search Screen.

Purpose of the Screen

This screen creates a simple first-time experience and provides a clear entry point into the application.

Good Morning, Student!

Ready for your classes today?



 Where do you want to go?

CURRENT LOCATION

 **Main Gate**



Quick Access



Classrooms



Laboratories



Offices



Facilities

Recent Locations

[View All](#)



Computer Science Block
North Campus



Central Library
Main Quad



Home



Search



Map



Profile

Figure 1: CampusNav Welcome Screen

SCREEN 2: HOME / SEARCH SCREEN

The Home Screen acts as the main dashboard of the Campus Navigation application. It allows users to search for destinations and quickly access frequently used campus facilities.

Main UI Elements

- Greeting message.
- Search bar.
- Current location.
- Facility categories.
- Campus map shortcut.
- Recent destinations.

Sample Interface Content

Good Morning!

Where do you want to go?

Search for:

Classroom, Lab, Office, Library...

Facility Categories

- Classrooms
- Laboratories
- Departments
- Library
- Cafeteria
- Offices
- Parking
- Restrooms

Purpose

The Home Screen provides users with immediate access to the most important navigation functions.

Instead of requiring users to browse multiple menus, the application places the search bar and facility categories directly on the main screen.

User Interaction

Users can type a destination into the search bar or select a facility category to browse available locations.

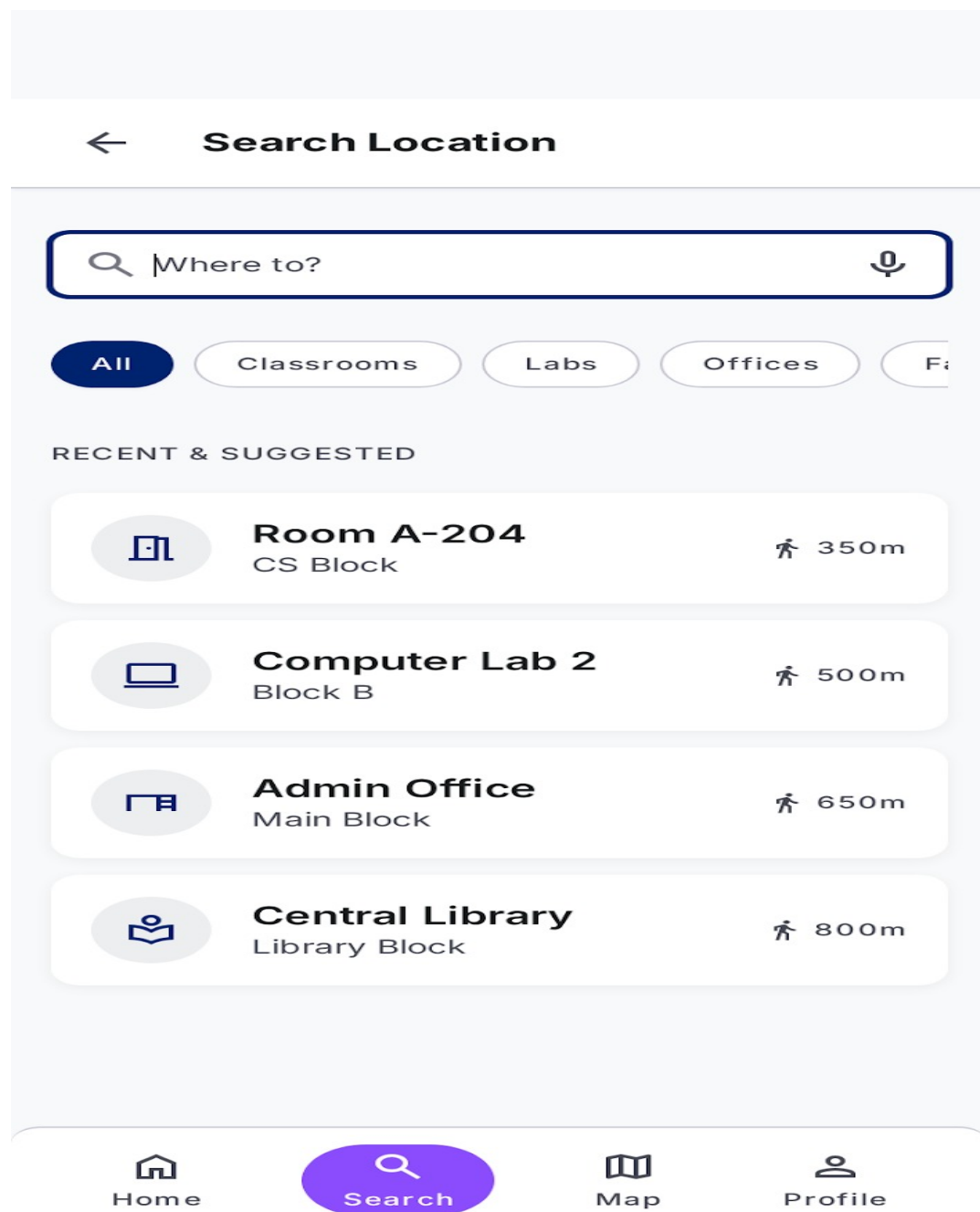


Figure 2:

CampusNav Home / Search Screen

SCREEN 3: CAMPUS MAP

The Campus Map Screen provides a visual representation of the campus and its major facilities. It helps users understand the physical arrangement of buildings, pathways, and important locations.

Main UI Elements

- Campus buildings.
- Roads and pathways.
- Current-location marker.
- Destination markers.
- Search option.
- Zoom controls.
- Navigation button.

Example Campus Locations

- Main Block.
- IT Department.
- Computer Laboratory.
- Central Library.
- Administrative Office.
- Cafeteria.
- Parking Area.
- Auditorium.

Purpose

The map provides users with a visual understanding of the campus layout and allows them to identify the relative position of different facilities.

User Interaction

Users can:

- Zoom in and out of the map.

- Select a building.
- Search for a location.
- View location information.
- Start navigation to a selected destination.

Design Consideration

Important buildings and facilities should be represented using clear labels and recognizable icons. The user's current position and selected destination should be visually distinguishable.

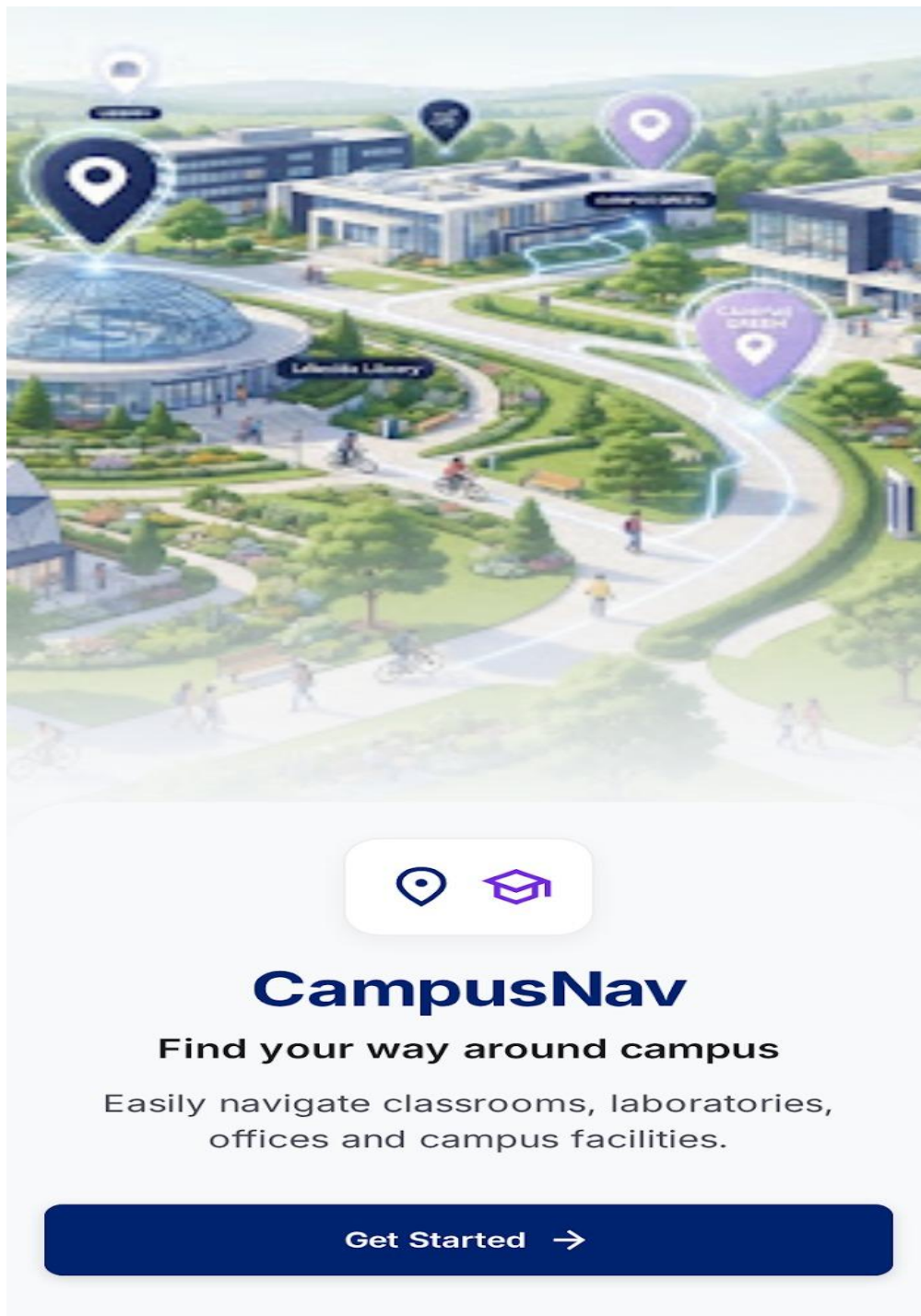


Figure 3: Interactive Campus Map Screen

SCREEN 4: SEARCH RESULTS AND LOCATION DETAILS

When the user searches for a location, the application displays relevant search results.

Example Search

Search Query: Computer Laboratory

Example Search Result

Computer Laboratory – Block A

Building: Block A

Floor: 2nd Floor

Distance: 250 m

Estimated Walking Time: 4 minutes

VIEW ON MAP

START NAVIGATION

After selecting a search result, the user can view additional information about the destination.

Information Displayed

- Facility name.
- Building name.
- Floor number.
- Distance from current location.
- Estimated walking time.
- Opening information, where applicable.
- Location on the campus map.

Purpose

This screen allows the user to verify the destination before beginning navigation. Providing the building, floor, distance, and estimated walking time reduces the possibility of selecting an incorrect location.

User Interaction

The user can select View on Map to see the destination or select Start Navigation to begin receiving directions.

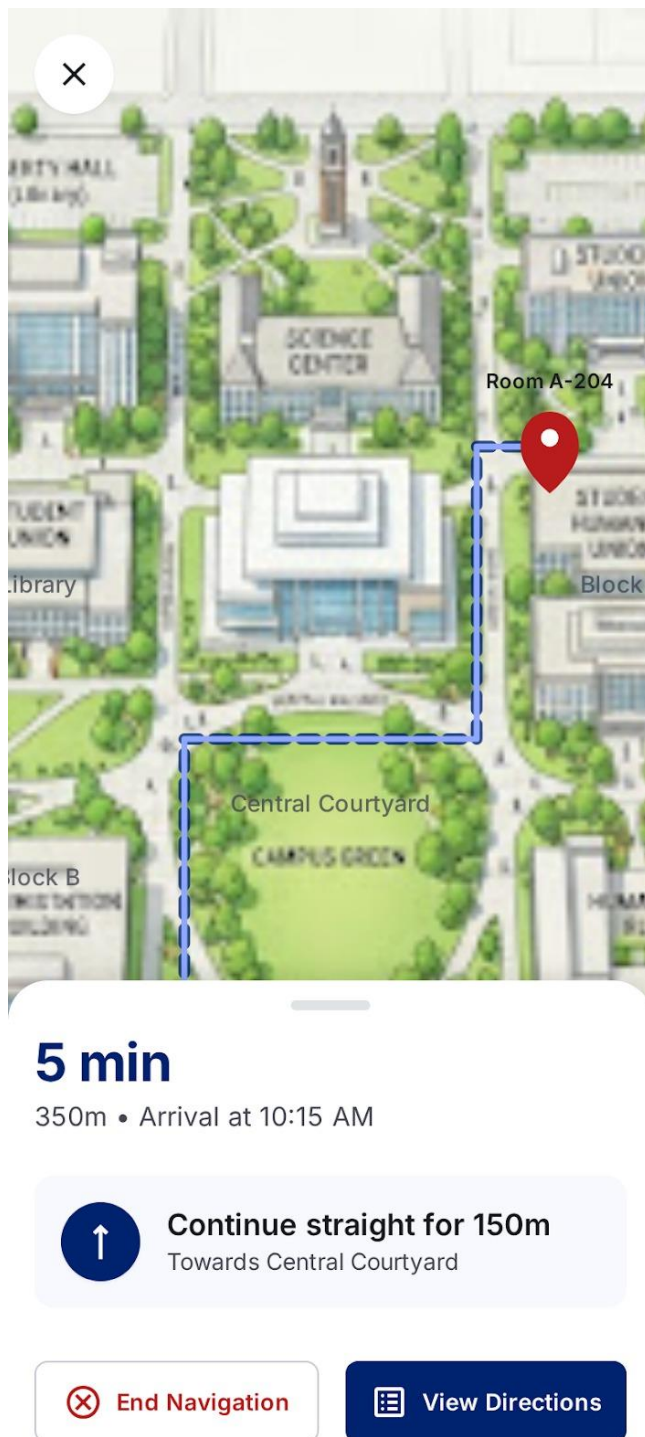


Figure 4: Search Results and Location Details Screen

SCREEN 5: NAVIGATION SCREEN

The Navigation Screen provides step-by-step guidance from the user's current location to the selected destination.

It is the final major interaction in the proposed application.

Main UI Elements

- Current location marker.
- Destination marker.
- Highlighted route.
- Distance remaining.
- Estimated walking time.
- Direction instructions.
- End Navigation button.

Example Navigation Information

Navigate to Computer Laboratory

250 m • 4 min

Directions

1. Walk straight from the Main Entrance.
2. Turn right at Block A.
3. Enter Block A.
4. Take the stairs to the second floor.
5. The Computer Laboratory is on the left.

Purpose

The Navigation Screen helps users reach their selected destination by providing clear route visualization and simple step-by-step instructions.

Proposed Navigation Features

The application can support:

- Walking navigation.
- Current-location tracking.

- Route visualization.
- Distance calculation.
- Estimated travel time.
- Turn-by-turn instructions.

User Interaction

The user follows the displayed route and instructions. After reaching the destination, the user can select End Navigation.

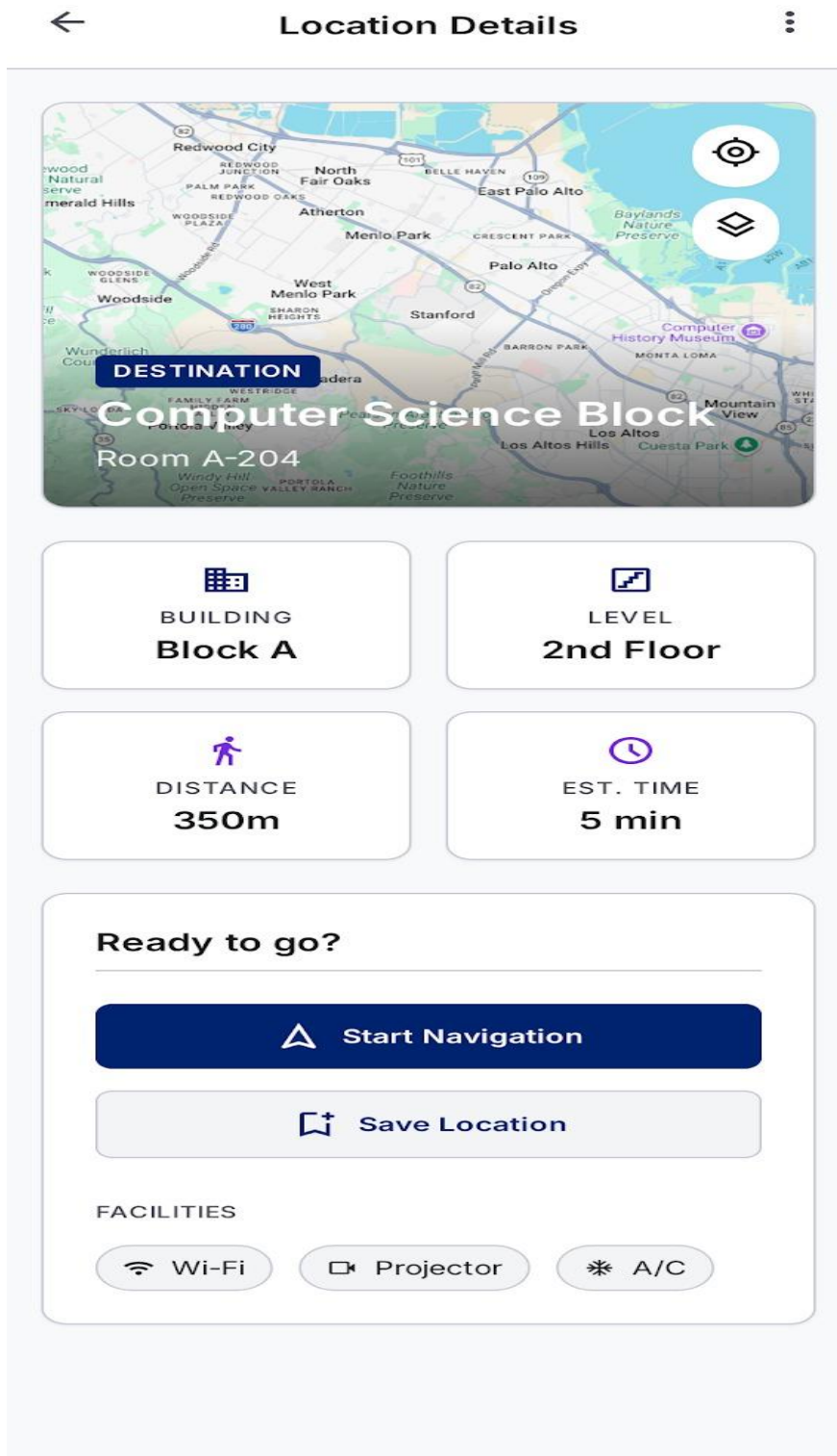


Figure 5: Campus Navigation and Route Guidance Screen

CONCLUSION AND FUTURE ENHANCEMENTS

The Campus Navigation Mobile Application provides a simple and effective solution for helping students, faculty members, and visitors navigate a large college campus.

The proposed UI focuses on usability, simplicity, consistency, accessibility, and efficient interaction. The application provides essential features such as location search, facility categories, campus map visualization, location details, route information, and navigation guidance.

The five primary UI screens demonstrate the complete navigation experience:

1. Welcome Screen.
2. Home / Search Screen.
3. Campus Map Screen.
4. Search Results and Location Details Screen.
5. Navigation Screen.

The proposed design provides a clear user journey from searching for a location to reaching the destination. With further development and integration with GPS, mapping, and indoor positioning technologies, the prototype can be transformed into a fully functional campus navigation application.